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Validity and reliability of the Bristol Stool Form Scale in healthy adults and patients with diarrhoea-predominant irritable bowel syndrome

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SUMMARY

Background
The Bristol Stool Form Scale (BSFS) is a 7-point scale used extensively in clinical practice and research for stool form measurement, which has undergone limited validity and reliability testing.

Aim
To determine the validity and reliability of the BSFS in measuring stool form in healthy adults and patients with diarrhoea-predominant irritable bowel syndrome (IBS-D).

Methods
One hundred and sixty-nine healthy volunteers provided a stool sample and used the BSFS to classify stool form, which was compared with measured stool water content and with values from 19 patients with IBS-D. Eighty-six volunteers used the BSFS to classify 26 stool models to determine accuracy and reliability.

Results
Volunteers' classifications of stool type correlated with stool water (Spearman's rho = 0.491, $P < 0.001$), which increased in hard (Types 1–2), normal (Types 3–5) and loose stools (Types 6–7) ($P < 0.001$). The BSFS detected differences in stool form between healthy volunteers (mean 3.7, s.d. 1.5) and IBS-D patients (mean 5.0, s.d. 1.2) ($P < 0.001$). Overall, 977/1204 (81%) stool models were correctly classified (substantial accuracy, $\kappa = 0.78$), although <80% of Types 2, 3, 5 and 6 were classified correctly. On 852/1118 (76%) occasions, volunteers classified covert duplicate models to the same stool type (substantial reliability, $\kappa = 0.72$), but with only moderate reliability for Types 2 (63%, $\kappa = 0.57$) and 3 (62%, $\kappa = 0.55$).

Conclusions
The BSFS demonstrated substantial validity and reliability, although difficulties arose around clinical decision points (Types 2, 3, 5, 6) that warrant investigation in larger clinical populations. Potential for improving validity and reliability through modifications to the BSFS or training in its use should be explored.

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INTRODUCTION

Stool consistency is a central component in the description of normal or altered bowel habit. Stool consistency generally refers to the rheology or viscosity of the stool[1, 2] and is determined largely by stool water content.[3] Rapid intestinal transit limits gastrointestinal water absorption leading to loose or liquid stools,[4] whereas slow intestinal transit results in extensive water absorption leading to harder stools.[5] The direct assessment of stool consistency involves measurement of stool rheology or viscosity, for example, using a penetrometer[1] or viscometer.[2]

The direct measurement of stool consistency or stool water content requires laboratory analysis and is impractical for routine purposes. Stool form can be considered as a proxy measure for stool consistency and refers to the shape and apparent texture of the stool which can be assessed visually.[6] Stool form scales are a standardised and inexpensive method of classifying stool form into a finite number of categories that can be used by doctors, health care professionals, researchers, patients and the general population, even in the absence of training.

Although a number of stool form scales have been developed and validated for use in adults[1, 6–8] one of the most widely used is the Bristol Stool Form Scale (BSFS).[9] The BSFS is an ordinal scale of stool types ranging from the hardest (Type 1) to the softest (Type 7). Types 1 and 2 are considered to be abnormally hard stools (and in conjunction with other symptoms indicative of constipation) while Types 6 and 7 are considered abnormally loose/liquid stools (and in conjunction with other symptoms indicative of diarrhoea). Type 3, 4 and 5 are therefore generally considered to be the most 'normal' stool form[10] and are the modal stool forms in cross-sectional surveys of healthy adults.[11] Although originally developed with written descriptors alone, pictorial representations of each stool type were later added for improved clarity (Figure 1).

The BSFS is widely used throughout the world in both clinical and research settings. For example, the Rome Foundation recommends use of the BSFS for sub-typing irritable bowel syndrome (IBS) according to the new Rome IV criteria[12] and the United States' Food and Drug Administration recommends the BSFS be used when assessing abnormal defaecation as the primary endpoint of trials of diarrhoea-predominant IBS (IBS-D).[13] The BSFS has also been used in a variety of research interventions and observational studies to estimate gut transit time (GTT).[4, 11, 14–19] It has been adapted into a number of languages[20–22] and has been modified for use in children.[23, 24] Despite its widespread use throughout the world, the BSFS has not undergone extensive validation.

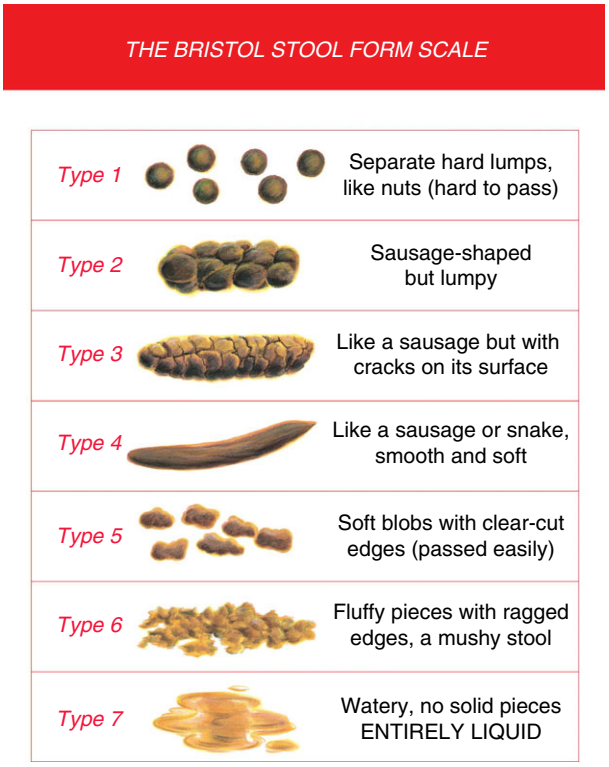

Figure 1 | The Bristol Stool Form Scale. Reproduced by kind permission of the late Dr K W Heaton, Reader in Medicine at the University of Bristol. © 2000 Norgine Pharmaceuticals Ltd.

Research findings and clinical decision-making are based upon the assumption that instruments are valid (accurate) and reliable (repeatable). The ability of the scale to correlate with a gold standard measure is key to demonstrating validity (concurrent validity).[25] Only one known study has assessed concurrent validity of the BSFS, which showed that stool type classified by healthy females was significantly correlated with actual stool hardness (measured with a penetrometer) and actual stool dry matter (measured by oven drying).[3]

A scale must also be able to detect clinically meaningful differences in the measured construct (construct validity).[26] For example, stool consistency and stool water are inversely related to GTT and there is fair evidence of a negative relationship between stool type measured using the BSFS and whole or segmental GTT in healthy adults,[9, 27] patients with IBS[4, 5, 14, 28] and

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patients with constipation.[14, 16] Furthermore, secondary analysis studies have shown that the BSFS score (mean BSFS stool type from prospective stool diaries) is different between healthy subjects and patients with IBS-D[29] and patients with chronic constipation.[16]

Moreover, unreliable instruments are inevitably invalid.[30] To date, there is no published literature demonstrating people classifying stools of the same stool form to the same BSFS type (i.e. intra-rater reliability).[25] This may be related to the feasibility of performing reliability studies that involve test–retest on real stools. Stool photographs have been used to assess reliability of the BSFS [12] and the modified paediatric BSFS,[24, 31] but assessing stool form involves shape and texture which are difficult to assess in two dimensional photographs.

The BSFS is widely used in clinical practice and research worldwide, yet there is a lack of dedicated validation studies of the BSFS and aspects of its reliability have not been demonstrated. The aim of this study was therefore to extensively assess validity and reliability of the BSFS in healthy adults and patients with IBS-D.

MATERIALS AND METHODS

This study involved the measurement of reliability and concurrent validity of the BSFS using stool models among 86 healthy volunteers and measurement of concurrent validity and construct validity using real stools among 169 healthy volunteers and 19 patients with IBS-D.

Participants

Healthy adults were recruited through email alerts sent to university staff and students, advertisements on volunteer recruitment websites, posters in the university and the snowball technique. The inclusion criteria for healthy volunteers were adults aged 18 years and over with a 'normal' self-reported stool frequency (between three bowel movements per day and three per week[11]) and English as at least a second language. Exclusion criteria were any history of gastrointestinal (GI) disease, previous GI surgery (except appendectomy and cholecystectomy), functional bowel disorders [including diarrhoea, constipation, IBS by Rome III criteria[32]], any co-morbidity likely to affect GI transit time, cognitive impairment and blindness.

Data from patients with IBS-D were taken from an intervention trial performed in our centre.[33] Patients with IBS-D were recruited from the IBS clinic of a London hospital and inclusion criteria were males and females aged 18–65 years with a diagnosis of IBS using Rome III criteria[32] and who had Type 6 or 7 stools according to the BSFS on at least 2 days during a 7-day screening period. Exclusion criteria were any history of GI disease other than IBS; previous GI surgery (except appendectomy and cholecystectomy); changes to IBS medication in the 4 weeks prior to the study; prebiotic, probiotic, lactulose or bowel preparation use within 4 weeks prior to commencing the study; and pregnancy or lactation (as required of the original trial protocol).

Written informed consent was obtained from each participant prior to participation. Ethical approval was granted by King's College London Research Ethics Committee for recruitment and participation of healthy volunteers and by Guys Research Ethics Committee for patients with IBS-D.

(A) Classification of real stools: comparison with stool water (concurrent validity). Concurrent validity of the BSFS was measured by comparison of the measured stool water content of real stools classified to different stool types of both lay participant and expert ratings.

Firstly, 169 healthy volunteers passed a stool and immediately classified its stool form using the BSFS. A subgroup of 59 stools that had been classified by healthy volunteers were then also classified by one of three experts in gastrointestinal research (subject to their availability). The experts were blinded to the classification provided by the healthy volunteer.

Stool water content was then analysed using lyophilisation. Briefly, the stool was homogenised for four minutes in a stomacher (Lab-Blender 400; Seward, London, UK). Approximately 3–5 g was then weighed into preweighed sample tubes, in duplicate, and frozen at −20 °C until analysis. Samples were lyophilised at −45 °C using a freeze-drier (Lyolab A; LSL Secfroid SA, Aclens, Switzerland) for a minimum of 4 days, weighed and returned to the freeze-drier until constant weight (within 0.01 g). The percentage stool water content was determined using the equation: (dry stool weight/wet stool weight) × 100.

The correlation between stool water content and BSFS stool type was then determined using Spearman's correlation coefficient (r_s). Once stool water data had been checked for normality (Shapiro–Wilk test), values were compared between stools classified to different BSFS stool types (each of Types 1–7) and also between stools classified to 'diagnostic groups' (hard Types 1–2; normal Types 3–5; loose Types 6–7)[10, 32] using a one-way analysis of variance (ANOVA) with a Bonferroni *post hoc* correction for multiple comparisons.

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(B) Classification of real stools: healthy participants vs. IBS-D (construct validity). In order to test construct validity, the stool type of the 169 stools from healthy volunteers were compared with that of the first stool passed by the 19 patients with IBS-D. Each participant provided one stool sample for analysis. The difference in BSFS types between healthy volunteers and patients with IBS-D were compared using a Mann–Whitney *U* test.

(C) Classification of real stools: comparison with experts (concurrent validity). Agreement between healthy volunteers and experts on classification of real stools was measured on a subgroup of 59 stools that had been classified both by experts and healthy volunteers (from study A). Agreement was compared using percentage agreement and kappa, as described in the statistical analysis section.

Stool models

Concurrent validity and intra-rater reliability of the BSFS was measured among 86 healthy volunteers who classified reference stool models in duplicate. Stool models were made from pre-determined quantities of dried mashed potato, food dye and boiling water, using a previously described method.[8] Seven stool models were prepared to represent the seven stool types on the BSFS. However, as stool form and stool water content are continuous variables and not restricted to seven discrete values, six intermediate stool models were also prepared in order that a full range of forms of human stool output were represented.

In order that the stool models accurately represented the seven types on the BSFS and six intermediate types, 25 models were initially created with a range of water content (50–88%) based on experimental data.[3, 6] These were then evaluated by three experts in gastrointestinal research who independently selected the seven models that best represented the BSFS types (e.g. Type 2, 56% water; Type 3, 62% water) and those at the mid-point water content were used to represent the intermediate stool types (e.g. Type 2–3, 59% water).

The 26 stool models (seven BSFS stool types, six intermediate stool types, all in duplicate) were presented to each healthy volunteer one at a time in a random sequence (Microsoft Excel 2010; Random Number Generator). Volunteers were allowed 30 seconds to classify each stool model using a laminated BSFS with both written and pictorial descriptors (Figure 1). Participants were instructed on how to use the pictorial and word descriptors of the BSFS to classify the stools, assigning each a category from Type 1 to Type 7. Participants did not know details of the stool modelling process nor the presence of duplicate stool models.

During the 12-month recruitment period, the 26 stool models were made using a standard recipe (e.g. percentage water, colouring) and the same modelling techniques with continued reference to the BSFS and photographs of the original models to ensure that the stool models were highly similar in appearance for all healthy volunteers (Figure S1).

(D) Correct classification of stool models: concurrent validity (accuracy). The accuracy (concurrent validity) of the BSFS was determined by comparing participant classifications of stool models for 'typical' BSFS types to the 'gold standard' classifications of Types 1 to 7 as determined by gastrointestinal experts (see above) and diagnostic categories of 'hard', 'normal' and 'loose'.

(E) Agreement in classification of stool models: intra-rater reliability. Agreement within healthy volunteers' classifications of covert duplicate stool models (intra-rater reliability) was measured by comparing the classifications given by each participant for duplicate models.

Statistical analysis

Percentage agreement was determined between raters (e.g. between classifications of real stools by healthy volunteers and experts in study C, between classifications of stool models by healthy volunteers with gold standard in study D). If the same stool type was assigned this was described as an exact agreement / correct classification, and if either the same or within one stool type was assigned then this was described as an exact/adjacent agreement or being correct to within one stool type. Exact agreement was calculated as a percentage and as a kappa coefficient (κ) to adjust for chance agreement,[34] and the values compared with published standards.[35] Exact/adjacent agreement was calculated as a percentage and as a weighted kappa (κ_w),[36] to account for the varying degrees of agreement/disagreement.[37, 38]

In order to detect factors contributing to validity and reliability, the accuracy of classification (percentage of all models classified correctly) and the intra-rater reliability (percentage of pairs with exact agreement) were compared between various groups (e.g. genders, university education etc) using a Mann–Whitney *U* test or Fisher's Exact test as appropriate.

All data were analysed using SPSS for Windows (Version 20.0) and a $P < 0.05$ was considered statistically significant.

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RESULTS

Of the healthy volunteers (studies A–E), 169 adults were recruited and provided a real stool. Eighty-six classified the reference stool models (33 males, 53 females; mean age 35 years 8 months (s.d. 13 years 9 months), the majority of whom were employed (n = 42) or students (n = 35). Although 18 volunteers did not consider English to be their first language, only one was not fluent. Of the volunteers with IBS-D (study B only), 19 patients provided real stools [9 males, 10 females, mean age 34 years 4 months (s.d. 9 years 7 months)].

(A) Classification of real stools: comparison with stool water (concurrent validity)

In total, 169 healthy volunteers provided a stool sample and classified its form using the BSFS. The stool type was moderately correlated with measured stool water (Spearman's rho = 0.491, $P < 0.001$). Despite considerable overlap, there was a significant difference in stool water between stools classified to the seven different stool types ($P < 0.001$), showing higher water content in stools classified to higher stool types (Figure 2). Interestingly, in the subgroup of 59 stools that were also classified by experts, stronger correlation between stool type and stool water (Spearman's rho = 0.701, $P < 0.001$) was achieved than when these were classified by healthy volunteers.

Pairwise comparison with *post hoc* correction showed significant differences in water content between many stool types, with some notable exceptions with overlapping and wide 95% confidence intervals, for example, Type 2 69.1% water (95% CI 67.1–71.0), Type 3 69.7% (67.4–71.9) and Type 5 73.3% (70.7–75.8), which were not statistically significant different in water content.

Interestingly, stool water was significantly different between different BSFS 'diagnostic groups' increasing from hard stools (mean 68.5% s.d. 6.3), normal stools (72.5%, s.d. 6.2) to loose stools (78.7% s.d. 5.6) ($P < 0.001$). Pairwise comparison with *post hoc* correction showed significant differences in water content between each of these diagnostic groups ($P \leq 0.001$).

(B) Classification of real stools: healthy participants vs. IBS-D (construct validity)

The participant-assigned BSFS stool type was significantly higher for the 19 stools from patients with IBS-D [median 5, (IQR 2); mean 5.0, s.d. 1.2] than for the 169 stools from healthy volunteers [median 4, (IQR 3); mean 3.7, s.d. 1.5] ($P < 0.001$). Almost half of the stools from patients with IBS-D were classified as loose Type 6 or Type 7 (9/19, 47.4%) compared with very few stools from healthy volunteers (23/169, 13.6%) ($P < 0.001$).

(C) Classification of real stools: comparison with experts (concurrent validity)

Of the stools provided and classified by the 169 healthy volunteers, 59 were then classified by an expert in gastrointestinal research. As expected, none of these stools were classified as Type 7 by either healthy volunteers or

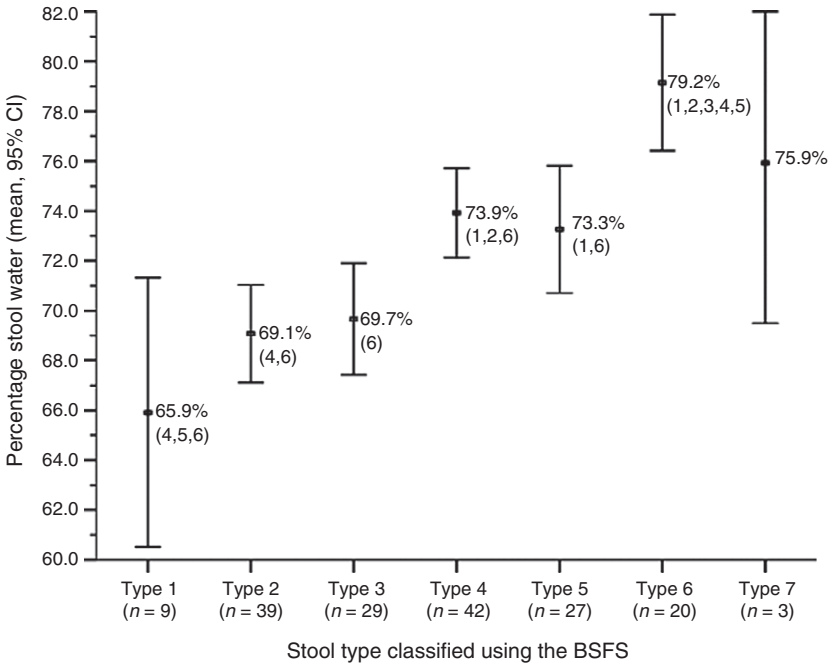

Figure 2 | Percentage stool water content of 169 stools classified using the Bristol Stool Form Scale by healthy volunteers. Data values are mean and error bars are 95% confidence interval of the mean. There was a significant difference in percentage water content of stools assigned to the seven stool types of the BSFS (ANOVA, $P < 0.001$). For each stool type, values in parentheses represent where significant differences exist following pairwise comparison with other stool types using a Bonferonni *post hoc* correction.

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the experts. Of the 59 stools, only 21 (36%) were correctly assigned to the same stool type classified by the experts, indicating only fair agreement ($\kappa = 0.25$) (Table 1). The percentage correctly assigned by healthy volunteers was not significantly different between genders, subjects familiar/unfamiliar with the BSFS, subjects with/without university degree (Fisher's Exact Test, all $P > 0.05$). However, 44 (75%) of the stools classified by healthy volunteers were correct to within one stool type of that classified by the experts, indicating substantial weighted agreement ($\kappa_w = 0.67$).

Stools classified as Type 2 by the experts were commonly classified by volunteers as Type 5 (5/12, 42%), while stools classified as Type 5 by healthy volunteers were commonly classified as Types 1–4 by experts (12/14, 86%) (Table 1). Interestingly only 36 (61%) of stools classified by healthy volunteers were in the correct 'diagnostics group' (i.e. classifying a Type 2 stool to a 'hard stool category', either Type 1 or Type 2).

(D) Correct classification of stool models: concurrent validity (accuracy)

Twenty-six stool models (seven matching the BSFS stool types and six intermediate types, all in duplicate, Figure S1) were classified by 86 healthy volunteers, representing 2,236 classifications in total.

For the models matching the seven BSFS stool types, 977/1204 (81%) were correctly classified, demonstrating substantial accuracy overall ($\kappa = 0.78$), while 1,144/1204 (95%) were correctly classified to within one stool type. Volunteers who were familiar with the BSFS correctly classified a greater proportion of the models than those who were not familiar with the scale ($P = 0.04$). Other demographic factors did not influence correct classification of stool models.

When analysed for individual stool types, the findings varied. For the models matching the seven BSFS stool types, correct classification ranged from 66% (Type 3) to 100% (Type 7), with near perfect classification of Type 1 and Type 7 models (Table 2). In contrast, Types 2, 3, 5 and 6 models were classified into a wide variety of stool types, with less than 80% being correct. Of note, at least 98% of classifications were correctly placed within one stool type, except for the Type 3 (80%) and Type 5 (90%) stool models (Table 2).

For the intermediate stool models, it was not possible to measure correct classification because the BSFS requires classification to one of seven discrete stool types and does not allow for intermediates. However, correct classification to adjacent stool types (e.g. classifying Intermediate 2–3 to either Type 2 or Type 3) ranged from 86% (Intermediate 1–2) to 98% (Intermediate 2–3) (Table 2).

(E) Agreement in classification of stool models: intra-rater reliability

As above, healthy volunteers classified 26 stool models, consisting of 13 stool models covertly presented in duplicate to measure intra-rater reliability. Overall, on 852/1,118 (76%) occasions volunteers classified the same BSFS stool type to duplicate stool models, representing substantial intra-rater reliability ($\kappa = 0.72$) (Table 3). However, for individual stool types, exact agreement ranged from 60% to 100%, with the lowest exact agreement for Type 2 (63%, $\kappa = 0.57$), Type 3 (62%, $\kappa = 0.55$) and intermediate Type 3–4 (60%, $\kappa = 0.54$), and the highest exact agreement for Type 1 (95%, $\kappa = 0.95$) and Type 7 (100%, $\kappa = 1.0$) stool models (Table 3). At least moderate intra-rater reliability was seen for each model ($\kappa \geq 0.41$).

Table 1 | Classification of real stools from 59 healthy volunteers compared to experts using the Bristol Stool Form Scale

	Stool type classified by healthy volunteers using BSFS						
Actual stool type (expert classification)	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7
Type 1	2	1	–	–	3	–	–
Type 2	–	5	2	–	5	–	–
Type 3	–	4	1	5	2	–	–
Type 4	–	2	3	4	2	1	–
Type 5	–	2	–	2	2	4	–
Type 6	–	–	–	–	–	7	–
Type 7	–	–	–	–	–	–	–

Dark grey shading denotes correct classification of real stools by healthy volunteers (based upon the experts' classifications). Dark grey plus light grey shading denotes correct classification to within one stool type. –, no classifications to this stool type.

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Table 2 | Classification of 26 stool models by 86 healthy volunteers using the Bristol Stool Form Scale

Actual Stool Type (stool models)	Stool type classified by healthy volunteers using BSFS: *n* (%)							Correct* *n* (%)	Correct to within one stool type† *n* (%)	Correct to diagnostic categories‡ *N* (%)
	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7			
Type 1	168 (98)	2 (1)	–	–	2 (1)	–	–	168 (98)	170 (99)	170 (99)
Intermediate 1–2	10 (6)	137 (80)	19 (11)	–	4 (2)	2 (1)	–	§	147 (86)	147 (85)
Type 2	5 (3)	128 (74)	36 (21)	–	3 (2)	–	–	128 (74)	169 (98)	133 (77)
Intermediate 2-3	2 (1)	24 (14)	142 (83)	2 (1)	2 (1)	–	–	§	166 (97)	§
Type 3	10 (6)	17 (10)	113 (66)	8 (5)	23 (13)	1 (1)	–	113 (66)	138 (80)	144 (84)
Intermediate 3–4	–	2 (1)	82 (48)	75 (44)	12 (7)	1 (1)	–	§	157 (91)	169 (98)
Type 4	–	2 (1)	18 (10)	138 (80)	14 (8)	–	–	138 (80)	170 (99)	170 (99)
Intermediate 4–5	–	–	6 (3)	165 (96)	–	1 (1)	–	§	165 (96)	171 (99)
Type 5	12 (7)	4 (2)	1 (1)	15 (9)	133 (77)	7 (4)	–	133 (77)	155 (90)	149 (87)
Intermediate 5–6	1 (1)	2 (1)	1 (1)	11 (6)	114 (66)	43 (25)	–	§	157 (91)	§
Type 6	–	1 (1)	–	1 (1)	42 (24)	125 (73)	3 (2)	125 (73)	170 (99)	128 (74)
Intermediate 6–7	–	–	–	–	11 (6)	117 (68)	44 (26)	§	161 (94)	161 (94)
Type 7	–	–	–	–	–	–	172 (100)	172 (100)	172 (100)	172 (100)

Twenty-six stool models (seven BSFS stool types, six intermediate stool types, all in duplicate) were classified by 86 healthy volunteers, resulting in 172 observations per stool type and 2,236 observations overall. –, no classifications to this stool type.

* This column denotes 'correct classification' of stool models by healthy volunteers (e.g. classifying a Type 2 model as Type 2). It is not possible to calculate 'correct classification' for intermediate stool models as the BSFS does not allow classification to intermediate stool types.

† This column denotes 'correct to within one stool type' (e.g. classifying a Type 2 model as Type 1, Type 2 or Type 3). For intermediate stool models, 'correct to within one stool type' is taken to mean the adjacent categories (e.g. classifying Intermediate 2–3 model to either Type 2 or Type 3).

‡ This column denotes 'correct to diagnostic category' (e.g. classifying a Type 2 model to a 'hard' stool category, that is, Type 1 or Type 2). It is not possible to calculate 'correct to diagnostic category' for intermediate stool models that cross diagnostic boundaries (i.e. Intermediate 2–3 and Intermediate 5–6).

§ not possible to calculate the numbers of correct classifications for this stool model.

Table 3 | Intra-rater agreement for 13 pairs of stool models classified by 86 healthy volunteers using the Bristol Stool Form Scale

Stool model (BSFS Type)	Exact agreement *n* (%)	Kappa	Exact and adjacent agreement, *n* (%)	Weighted kappa
Type 1	82 (95)	0.95	84 (98)	0.95
Intermediate 1–2	62 (72)	0.67	80 (93)	0.88
Type 2	54 (63)	0.57	81 (94)	0.92
Intermediate 2–3	72 (84)	0.81	82 (95)	0.96
Type 3	53 (62)	0.55	64 (74)	0.80
Intermediate 3–4	52 (60)	0.54	83 (97)	0.93
Type 4	62 (72)	0.67	84 (98)	0.96
Intermediate 4–5	79 (92)	0.91	85 (99)	0.99
Type 5	64 (74)	0.70	75 (87)	0.78
Intermediate 5–6	57 (66)	0.61	83 (97)	0.91
Type 6	69 (80)	0.77	84 (98)	0.95
Intermediate 6–7	60 (70)	0.65	85 (99)	0.96
Type 7	86 (100)	1.00	86 (100)	1.00

Familiarity with the BSFS, gender, a university education and health care experience did not influence the percentage of pairs of stool models for which volunteers demonstrated exact agreement (all $P > 0.05$).

Overall, exact and adjacent agreement (agreement to within one stool type) was 1056/1124 (94%) with a $\kappa_w = 0.90$. Only two models achieved less than 90% exact and adjacent agreement, for Type 3 (74%,

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κ_w =0.80) and Type 5 (87%, κ_w = 0.78) stool models (Table 3).

DISCUSSION

The BSFS is widely used throughout the world in both clinical and research settings, but has not undergone extensive validation. Overall the BSFS demonstrated concurrent validity and construct validity, and showed substantial reliability.

There was correct classification of the majority of stool models and different stool water content was found in real stools classified to different BSFS stool types, both of which indicate concurrent validity. However, not all aspects of this type of validity were consistently positive. Firstly, only fair exact agreement was found between healthy volunteers and experts in the classification of real stools. Secondly, although real stools across the whole BSFS had different water content, there were minimal differences between individual stool types and there was wide overlap in water content of stools classified to different stool types (Figure 2). In practice, this would mean that a stool of specific water content (e.g. 71%) might be classified into a wide range of stool types. Stool hardness[39] and stool form[2] have previously been found to have a nonlinear relationship with stool water content. Thirdly, the correlation between stool water and healthy volunteers' classification of BSFS stool type was relatively low (Spearman's rho = 0.491). In contrast, when classified by experts, the correlation was much higher (Spearman's rho = 0.701), justifying the use of experts as a 'gold standard' reference and suggesting that frequent, routine use of the BSFS may result in greater validity.

Other stool form scales have been able to show differences in stool water content between each stool type[6, 7] and have reported much higher correlation between stool water and stool type (Spearman's rho = 0.78).[40] However, these scales used only four stool form categories. The BSFS has seven categories and therefore allows differentiation between small variations in stool form but correct classification is inevitably lower when users are faced with greater number of categories from which to select.[10] Similarly, the widest ranges of water contents were found at the extremes of the BSFS (Types 1 and 7), where participants could not choose a 'harder' or 'looser' alternative. Such 'ceiling' and 'floor' effects have been documented with other gastrointestinal function scales.[12]

The accuracy of classification of reference stool models was greater than the classification of real stools. A number of factors may affect accurate classification of real stools, including multiple stool forms within the same toileting episode[41] or even the within the same stool.[39] Healthy volunteers and experts may have varied in how they classified a stool of mixed form, for example, by classifying either the most predominant type or a mid-point between two types. However, such variation in stool form did not occur with stool models, likely artificially improving accuracy of their classification. Stool size and shape may also impact the accuracy of classification because smaller stools may lack the characteristic features that enable their categorisation. For example, Type 1 and Type 5 are the only stool types pictured as small individual pieces, and therefore a small stool of another form may be misclassified as being Type 1 or 5. In addition, these are the only stool types that include description of the sensation on passing (Type 1 'hard to pass'; Type 5 'passed easily'), which may aid classification when used by the person passing the stool (as in this study) but not when used by an external rater (e.g. a nurse or carer or our expert raters). Type 1 stools that are passed without difficultly may be misclassified by the person passing the stool as Type 5, and this occurred in this study (Table 1). Gastrointestinal disorders such as IBS-C or IBS-D may further skew perception of stool form, due to the accompanying symptoms experienced, such as urgency or pain on defaecation.

In both classification of real stools and stool models, correct classification to within one stool was inevitably higher than correct classification alone. Whether being correct to within one stool type is sufficiently acceptable for clinical and research purposes depends upon the context in which the BSFS is being used and indeed where the discrepancies occur. Generally, Types 3, 4 and 5 are considered 'normal' stool form[12] and therefore stools at the borders of 'normal' form (i.e. Type 2 and Type 3; Type 5 and Type 6) may require exact classification to distinguish between normal and abnormal bowel function. However, in the current study, Types 2, 3, 5 and 6 actually had the lowest exact correct classification of both stool models and real stools, which may result in patients being miscategorised as having normal or abnormal stool form. This confirms a recent study of gastroenterologists using the BSFS to classify stool photographs that showed diverging classifications of stools around these important IBS-diagnosis cut-off points.[12] While the findings from the small sample of IBS-D patients in this study should be considered preliminary, they may suggest implications for IBS clinical management and research as the Rome Foundation recommends the use of the BSFS for sub-typing[12] and the United

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States' Food and Drug Administration recommends its use as a primary endpoint of trials of IBS-D.[13]

Overall, substantial intra-rater reliability of the BSFS was demonstrated in the classification of stool models (overall exact agreement 76%). The highest agreement was at the extremes of the scale, which may be explained by volunteers only having one direction in which they could alter their classifications. It might have been expected that intra-rater reliability would be low for the intermediate stools that fell between two types points on the scale, however, five of six models had substantial reliability or higher. Agreement was similar to that reported with other stool charts, including those with fewer stool type categories (72% for a four category stool chart).[42] The high agreement seen in the current study may have been partly influenced by participants seeing all the models within a short period and therefore experiencing a 'training effect'. Indeed, patients may improve the accuracy of classifications over time through increasing familiarity with scale for clinical purposes.

Harnessing this training effect may be one way to improve the accuracy of lay classifications. Patients could be provided with more detailed instructions, for example, highlighting common errors made in classifications. Otherwise they could be allowed to practice classifications of stool models or photographs under supervision to improve accuracy prior to commencing a clinical trial.

An alternative approach to improving the accuracy of the scale would be to modify it, perhaps by reducing the number of stool types or by changing their descriptors. The modified Bristol Stool Form Scale for Children (mBSFS-C) has the same text and pictorial descriptors as the BSFS, but does not include types 3 and 5 from the original scale.[31] The mBSFS-C has already shown good concurrent validity and inter-rater reliability in estimating stool form amongst children and paediatric gastroenterologists.[23, 24, 31] Decreasing the number of categories increases the ability to discriminate between groups which may be sufficient in a clinical setting [43] but reduces potential responsiveness of the scale (sensitivity to change) which is important when considering using the BSFS as a clinical trial endpoint.[13] Alternatively, the text or pictorial descriptors of the original BSFS could be adapted to improve validity and reliability. Limited agreement between the words used by patients to describe the BSFS pictures and the text descriptors has been described[44] that may partly account for the confusion between Types 1 and 5, both of which are depicted as separate lumps. Rephrasing the text descriptors for each picture using words commonly used by laypersons could further improve the performance of the scale, particularly for BSFS types 2, 3, 5 and 6, which showed the poorest concurrent validity with expert classifications. However, any modifications to the BSFS would require further validation to measure its impact on accuracy.

STRENGTHS AND LIMITATIONS

The BSFS was validated by comparing stool types classified by participants to objective measures of stool consistency including water content of fresh stool samples and expert classifications. This study also used stool models which more closely reflect real stools than do photographs that have previously been used in validation studies[24, 31] while limiting the biohazard of exposing participants to a range of real stools. The use of stool models also allowed the selection of intermediate stools representing the full range of possible stool output.[45] The study included both healthy volunteers and patients with IBS-D, although only the healthy volunteers contributed to the analysis of concurrent validity of real stools with stool water content. Measurement of the validity of the BSFS in other diseased populations and also in health professionals who may use it would be valuable given the varying perceptions of diarrhoea and constipation among these groups.[46, 47]

In this validation study, all recordings were prospective, the appearance of stool models was controlled and participants were asked to focus on accurate stool classification.[48] A review found that reliability was reduced by up to 35% in covert compared to overt reliability studies.[49] Therefore, the results reported here are likely to be inflated compared with what would be expected during routine use in the clinical and research settings.[50] However, this may be counteracted partly by the training of participants in the use of the BSFS, as it sometimes occurs in clinical or research settings.

As only one expert classified each real stool, inter-rater reliability between experts could not be determined. Investigation of the validity and reliability of the BSFS when used by health professionals who have frequent clinical exposure to it (e.g. gastroenterologists, nurses, health care assistants) is now warranted. The sample size of 19 patients with IBS-D was limited and based upon our findings, further validity and reliability testing, including the use of stool models, should be undertaken in larger clinical populations. Finally, the new Rome IV diagnostic criteria emphasise the importance of using the BSFS in the diagnosis of IBS, but also the need to measure stool output over 14 days.[12]

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CONCLUSION

Overall, the BSFS demonstrated substantial validity, demonstrated by comparison with stool water content, distinction of stools between healthy volunteers and patients with IBS-D and correct classification of the majority of real stools and stool models. In some cases, stool classification was problematic, with only fair agreement between healthy volunteers and experts. The validity of the BSFS for individual stool types was variable and difficulties arose predominantly around important clinical decision points on the scale (Types 2, 3, 5 and 6), although significant differences in water content were found between hard, normal and loose stool classifications. Familiarity with the scale and education in its use may partly address these issues.

To our knowledge this is the first study to measure the reliability of the BSFS in healthy volunteers, showing substantial intra-rater reliability. Further research is required to assess the concurrent validity and reliability in different patient groups, and in larger clinical populations, to make direct comparisons with other stool form scales. The potential for improving validity and reliability through modifications to the BSFS or the training of patients in its use should be further explored, and direct comparison with other methods would enable judgements regarding the ideal approach to recording stool output. The BSFS is already widely used to measure stool form in the clinical and research setting, despite limited psychometric data regarding its validity and reliability. Rather than providing blanket support for use of the BSFS in such ways, these data highlight aspects of the scale that are strong, as well as those that clinicians and researchers should treat with healthy caution.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Figure S1. Stool models representing the seven stool types of the BSFS and intermediate stool types, presented in duplicate to healthy volunteers.

AUTHORSHIP

Guarantor of the article: Prof Kevin Whelan.

Author contributions: MB and KW designed the study. MB and JR collected and analysed data. All authors (MB, KW, JR) interpreted the data, assisted in writing the paper and approved the final version of the article.

All authors approved the final version of the manuscript.

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